

Proteus report

An H-bridge motor driver is an electronic circuit that allows a DC motor to be driven in both clockwise and counter-clockwise directions. By controlling the switching of four transistors in an H-shaped configuration, the polarity of the voltage applied to the motor is reversed, enabling bidirectional control

Components used :

- Transistors

To allow the current flow , 2 transistors work at same time to allow flow of in one direction then other 2 transistors work and which reverses the polarity and drives motor in opposite direction

- Resistors

Connected between switch and transistors to limit the current in base to avoid damage of transistors

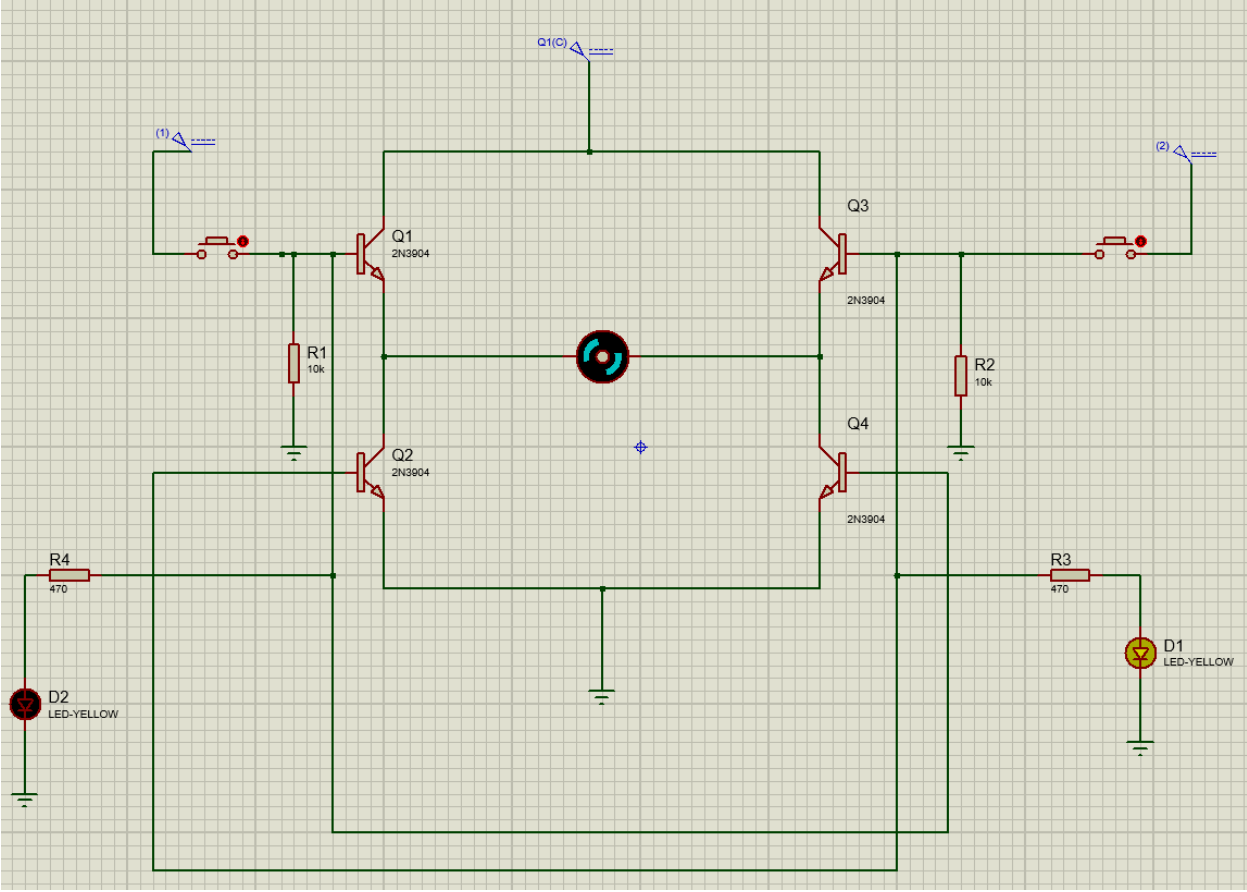
Other resistors was used in series with leds to avoid burning out

- LEDs

Used as an indication for motor direction

- Switches

Two switches are provided for **manual control** of the motor direction



LTspice report

Circuit 1 : PNP

Components used :

Zener, pnp transistor , resistors ,voltage source

At low voltage in my case which is less than 9v the Zener diode does not conduct so Q1 base gets no drive which results Q1 and Q2 remain OFF, and the load sees no current but when input rises above 9.1 v the Zener breaks down and starts conducting and the Current flows through R1 to zener to base of Q1 which results that Q1 turns ON and drives Q2 into conduction , Q2 acts as the main power switch to the load (R2, R4) gets current.

Circuit 2 : NPN

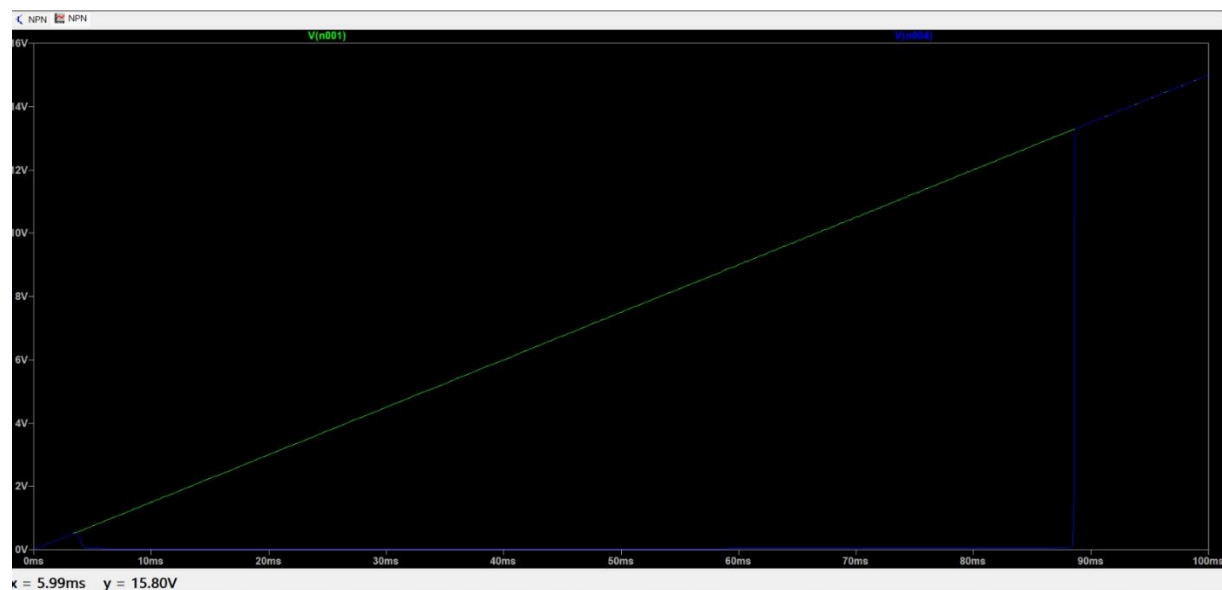
Components used :

nnp transistor , resistors ,voltage source

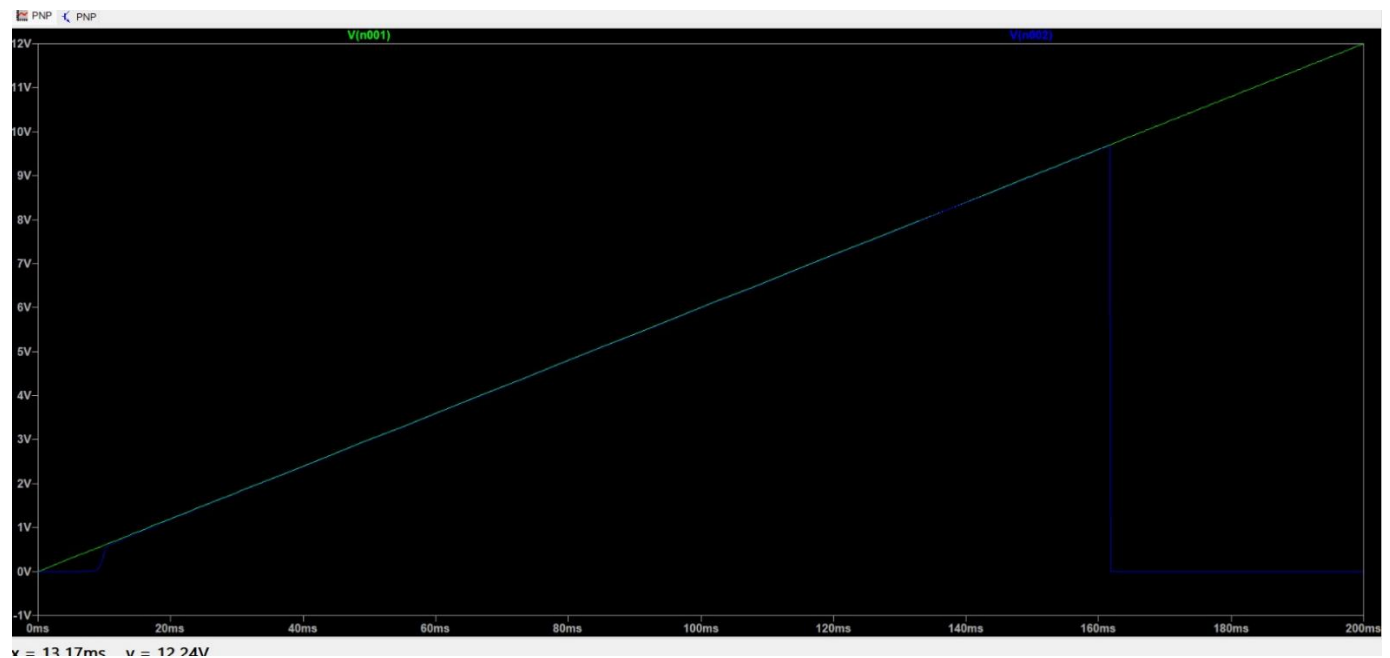
At low voltage the divider output is too small $< 0.7V$ to turn Q1 ON. so both Q1 and Q2 stay off and the load is disconnected as input increases, the divider output rises

Q1 on results to drives Q2 then Q2 switches on and allows current through the load R4

NPN



PNP



- Vin (green line) increases linearly from 0V to 12V.
- Vout (blue line) stays low initially, then jumps high at around 160ms.
- After that, it stays high until the end of the simulation.
- This suggests a threshold behavior, where the transistor starts conducting or switching when Vin reaches a certain point

Other

- Vin (green line) increases linearly again, this time up to ~15.8V.
- Vout (blue line) remains high and then drops sharply low at around 90ms.
- This indicates that the NPN transistor switched on, pulling the output low once Vin exceeded a certain threshold.

The main difference is that the PNP transistor circuit causes the output to rise when Vin reaches a certain level while the NPN transistor circuit causes the output to drop when Vin reaches the threshold

